

# Owned Single-Wire Transactions

Prove a bounded policy before touching a bus

**E0 SYNTHETIC REPLAY — NO PIN, PULL-UP, OR SENSOR**

**Lesson** 064 — component  
**Status** host verified and published at E0  
**Input** copied requests and line receipts

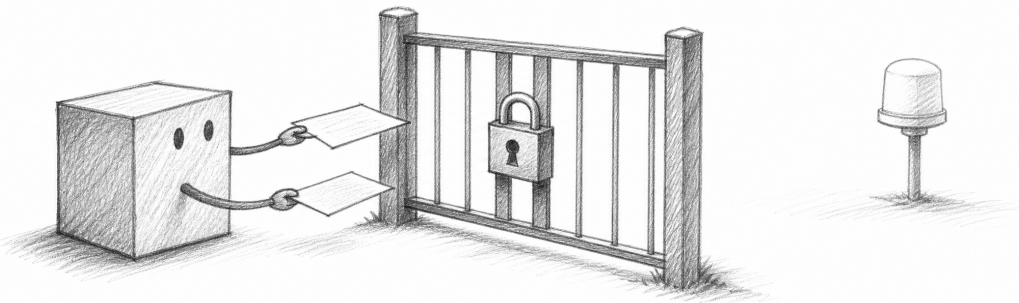
**Time** 180–240 minutes  
**Board** compile-only Mega replay  
**Observation** named intent/snapshot cells

**Responsibility.** `OneWireTransactionPolicy` expands one reviewed typed operation into a bounded sequence of copied line intents, validates one correlated copied receipt at a time, retains an attributable accepted prefix, and requests release during completion or rollback. It cannot drive, sample, power, or release a physical line.

## 1 Separate protocol policy from electrical execution

The DS18B20 data sheet describes an open-drain 1-Wire interface, reset and presence behavior, read and write slots, ROM commands, supply modes, and device timing limits.<sup>1</sup> Those facts explain what a future qualified adapter must implement. They do not prove that an unidentified kit part, a Mega pin, or a particular pull-up meets the requirements.

Lesson 064 therefore stops at a copied evidence boundary. The policy emits semantic values such as `DriveLow`, `Release`, and `Sample`. A synthetic producer returns a copied receipt saying what it applied or observed. E0 proves only that the policy processes that evidence deterministically.



*Alternative text: a graphite policy box hands two blank copied cards toward a locked gate; a faint generic sensor silhouette remains isolated beyond the gate, and no physical connection crosses it.*

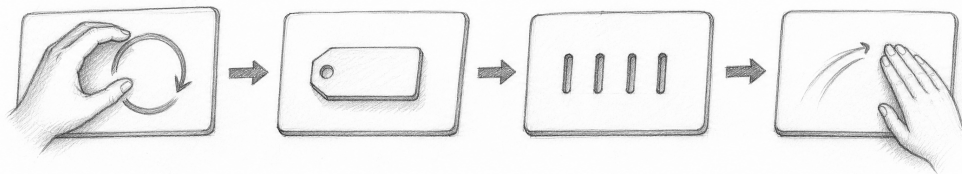
| E0 can establish                          | E0 cannot establish                                  |
|-------------------------------------------|------------------------------------------------------|
| typed phase order and receipt correlation | pin ownership, voltage, or open-drain behavior       |
| configured timing-window acceptance       | physical pulse width, rise time, or interrupt margin |
| bounded rollback and release intent       | a physical line reached idle                         |
| byte-stable replay from copied evidence   | genuine DS18B20 identity or thermal behavior         |

There is no formal electrical schematic or wiring exercise in this lesson. An authoritative schematic requires an exact specimen, package and pinout, ordinary pull-up and supply choice, endpoint ownership, measured rails and waveforms, and separate resource and safe-state evidence.

## 2 Use a closed operation vocabulary

The public request chooses one reviewed operation. Reset/presence, one bounded ROM-search pass, and single-drop ROM read cover identification mechanics. Addressed operations cover power-supply read, conversion start, one conversion status continuation, and scratchpad read. Their public enum spellings are `ResetPresence`, `SearchRomPass`, `ReadRomSingleDrop`, the four `MatchRom...` operations. No request accepts arbitrary command bytes, captured pulse trains, or an unknown replay script.

<sup>1</sup>Analog Devices, *DS18B20 Programmable Resolution 1-Wire Digital Thermometer*, Rev. 6.



*Alternative text: four graphite index cards progress through a reset gesture, an identity tag, bounded slot marks, and an open-hand release gesture; the surrounding prose supplies the operation names.*

Addressed operations reject only the all-zero ROM. Every other complete 64-bit value is transported byte-for-byte, including a wrong-family or bad-CRC value. Lesson 064 does not call it valid. Lesson 065 exclusively owns family-0x28 and Dallas CRC qualification.

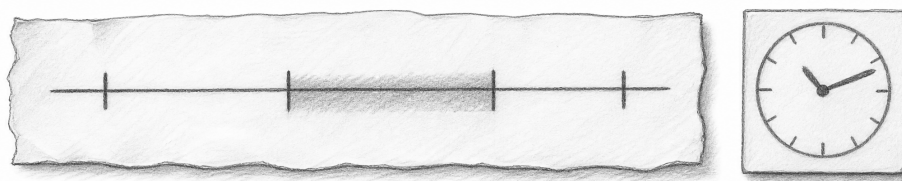
**ReadRomSingleDrop** is admitted only when configuration explicitly names a single-drop fixture. Multidrop work uses exactly one bounded **SearchRomPass**. The manufacturer search algorithm explains why each device's 64-bit ROM participates in a discrepancy tree,<sup>2</sup> but its enumeration order is evidence order, never physical position.

Every ordinary operation begins with reset and presence. **ResetPresence** stops there without sending a command. The sole reset exception is **MatchRomReadConversionStatus**: it is admitted only as a one-slot continuation of this policy's immediately preceding completed **MatchRomStartConversion** for the same nonzero ROM. A free-standing status request rejects atomically.

Why should a wrong-family nonzero ROM be transported rather than “fixed” by Lesson 064? \_\_\_\_\_.

### 3 Make microsecond time explicit

The repository's millisecond time type cannot represent single-wire slot windows. Lesson 064 uses a dedicated unsigned 32-bit **MicrosecondTimePoint** and **MicrosecondDuration**. Configuration names independent lower and upper limits for reset low, reset release, presence start, presence low, write-zero low, write-one low, read initiation, read sampling, complete slot, recovery, and the whole transaction deadline.



*Alternative text: a graphite timeline on torn paper has four ticks and a shaded admissible window between the middle ticks; a separate clock card represents caller-supplied microsecond time.*

Tests inspect immediately below, exactly at, and immediately above every lower and upper boundary. All configured durations stay strictly below the modular half range. Backward time and an exact half-range difference are ambiguous and reject atomically; an ordinary unsigned rollover with a valid forward distance is accepted.

Analog Devices' software guidance emphasizes that recovery and sample timing depend on the device and network conditions.<sup>3</sup> The configured E0 windows are copied-evidence acceptance rules, not a claim that any physical waveform met them.

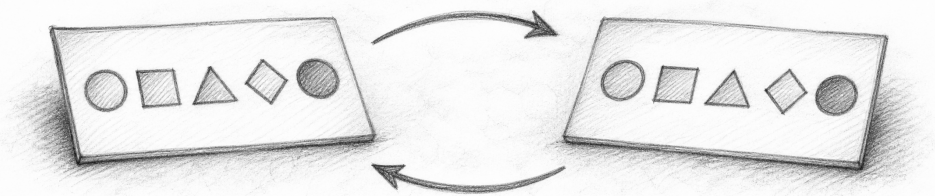
### 4 Correlate every intent and receipt

An operation request carries request sequence, typed operation, optional addressed ROM or fixed search state, supplied start time, supply mode, and producer status. Every emitted intent binds owner, lifecycle and configuration revisions, request and transaction generations, phase sequence, slot index, operation, line intent, legal action window, and addressed ROM.

<sup>2</sup>Analog Devices, *1-Wire Search Algorithm*, Application Note 187.

<sup>3</sup>Analog Devices, *1-Wire Extended Network Standard*, discussing Application Note 126 and network-dependent timing.

A receipt carries producer source and configuration, its own sequence and observation time, then repeats owner, lifecycle, request, transaction, operation, phase, phase sequence, slot, and applied-line correlation. It adds the copied sampled level, acceptance bit, and producer status. The addressed ROM and legal window remain in the owed intent and are checked through that binding; they are not receipt fields.



*Alternative text: two graphite cards face each other with the same ordered set of geometric seals and opposing arrows, representing exact field correlation and one return receipt.*

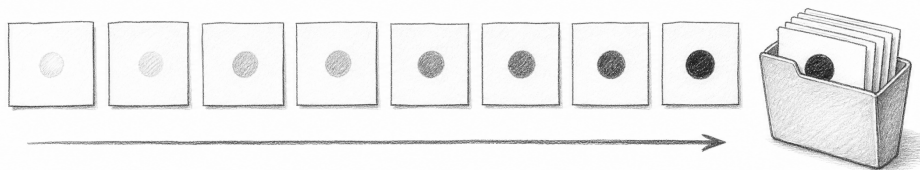
A matching receipt advances at most one phase or slot. A byte-identical duplicate is idempotent. Changed duplicates, crossed transactions, a foreign source, wrong generation, wrong operation, or wrong phase or slot reject atomically. A structurally matching receipt outside its inclusive window is retained as attributed timeout evidence and starts one release rollback; it does not advance the requested operation.

Sequence regression and half-range ambiguity reject atomically. If a call cannot leave a unique next transaction, phase, lifecycle, receipt, or cleanup correlation, the entire call returns **CapacityExceeded** without mutation. No counter wraps to zero.

1. If every field matches except the slot index, should the policy use the sampled bit? \_\_\_\_\_.
2. If a byte-identical receipt is presented twice, how many phase transitions occur in total? \_\_\_\_\_.
3. A receipt has valid correlation but producer failure. Does structure reject, or does rollback begin?  
\_\_\_\_\_.

## 5 Serialize least-significant bits first

The DS18B20 protocol sends data and commands least-significant bit first. Lesson 064 freezes the request at **begin()**, expands the allowlisted operation into bounded private slot mechanics, and assembles copied read bits back into bytes in the same order.



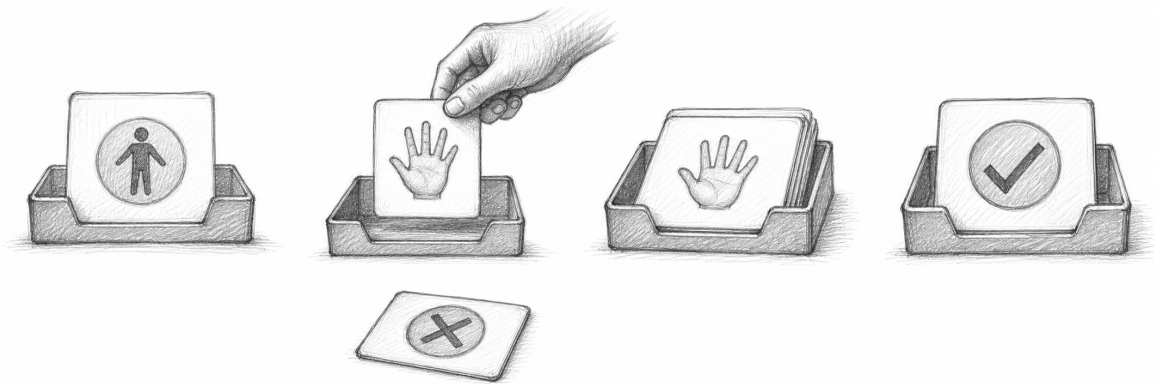
*Alternative text: eight graphite cards with progressively darkened dots run left to right into a folder, representing least-significant first slot order and an attributable accumulated prefix.*

Partial read bytes are prefix evidence, not a completed device response. A failure at the first, middle, or final slot preserves exactly the admitted prefix and moves to cleanup. Neither an incomplete scratchpad nor a copied search branch becomes a temperature.

For copied bits 1,0,1,1,0,0,0,1 in slot order, write the assembled byte in binary: \_\_\_\_\_. Which bit becomes the most significant bit? \_\_\_\_\_.

## 6 Treat release as an owed transaction

The policy never emits “drive high.” Its canonical inactive and cleanup line intent is **Release**. Initialization, reset, cancellation, timeout, producer failure, and shutdown may all leave one correlated release receipt owed. Exhaustion is the exception: if no unique cleanup correlation can be minted, the entire triggering call rejects atomically rather than promising an unconfirmable release.



*Alternative text: four separated graphite trays show active work, placement of an open-hand release card, a pending stack, and a confirmed card; a failed card lies separately below and does not overlap the path.*

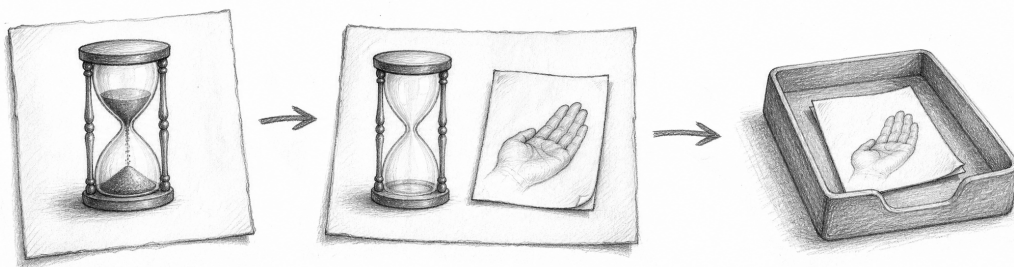
| Evidence                        | Permitted statement                             |
|---------------------------------|-------------------------------------------------|
| policy emitted <b>Release</b>   | software requested line release                 |
| matching copied release receipt | producer reported applying the request          |
| future E1 waveform observation  | qualified physical line reached its idle window |

**confirmCleanup()** is the sole closing seam. Reset becomes ready only after its matching copied release receipt. Shutdown remains initialized in a closing state so it can accept that receipt, then becomes inert. Shutdown supersedes any earlier pending cleanup with a fresh lifecycle and correlated **Release**; the earlier receipt becomes stale. Without the closing receipt, the snapshot remains **ReleaseUnconfirmed**; E0 never says the physical bus is idle.

**completedEvidence()** returns **ResourceBusy** after successful data slots while terminal release remains unconfirmed. It returns **Ok** with the complete attributed snapshot only after that receipt, and returns **NotInitialized** for every other state.

## 7 Advance deadlines without polling

Ordinary phase transitions require one receipt through **update()**. The receipt-free **advance(now,intent)** path does only one job: inspect supplied time. At or before the deadline it leaves the owed phase unchanged. Only the first supplied time strictly after the deadline emits at most one correlated release intent. It never polls, retries, catches up, invents phase evidence, or synthesizes a cleanup receipt.



*Alternative text: a graphite hourglass runs, then empties beside one open-hand release card, which remains in a tray as the same unresolved cleanup obligation on later calls.*

**MatchRomReadConversionStatus** is a one-slot continuation, not a free-standing poll transaction. It is available only after this policy completed **MatchRomStartConversion** for the same ROM, and it reports one copied bit. A later status check requires another completed conversion start; the component hides no poll loop.

Predict the result of calling **advance()** repeatedly after a timeout but before cleanup confirmation: number of successful phase receipts invented \_\_\_\_\_; cleanup quality \_\_\_\_\_.

## 8 Keep physical external-power evidence open

Only the copied **ExternallyPowered** supply-mode value is admitted at E0; admitting that enum does not energize or qualify a circuit. **ParasitePower** returns **Unsupported** before state changes. There is no strong-pull-up phase or intent. Ordinary **Release** must never be reinterpreted as output-high drive or strong-pull power.

The data sheet explains that parasite operation uses the data line for power and imposes additional strong-pull requirements during operations such as temperature conversion. That is precisely why it is excluded here. A future parasite-power endpoint needs separate topology, current, rail, timing, exclusivity, rollback, stuck-on handling, and independent power-removal evidence.

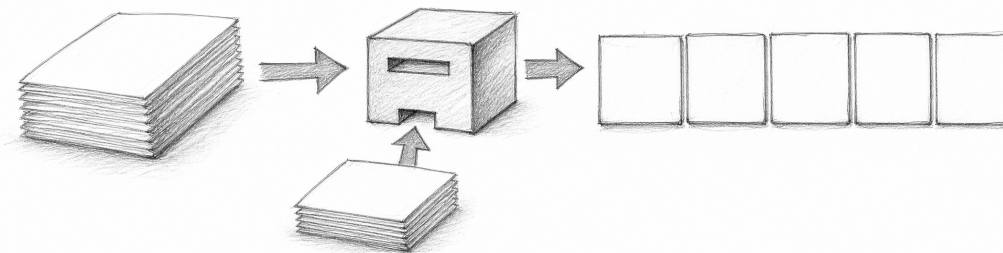
| Request                            | E0 disposition     | Reason                                  |
|------------------------------------|--------------------|-----------------------------------------|
| externally powered typed operation | eligible           | copied protocol policy only             |
| parasite-powered operation         | <b>Unsupported</b> | no reviewed power capability            |
| arbitrary captured transaction     | invalid            | outside closed operation surface        |
| scratchpad EEPROM copy             | absent             | endurance and strong-pull work excluded |

## 9 Replay the state machine

The compile-only Mega sketch supplies microsecond values, requests, receipts, and fixed caller-owned output buffers. It does not include a GPIO adapter and must not be uploaded as a bus driver.

```
OneWireStepIntent intent = {};
OneWireTransactionSnapshot snapshot = {};
OneWireTransactionSnapshot evidence = {};

OneWireTransactionPolicy transaction(config);
Status status = transaction.initialize(now, intent);
status = transaction.confirmCleanup(now, releaseReceipt);
status = transaction.begin(now, request, intent);
status = transaction.update(now, phaseReceipt, intent);
status = transaction.snapshot(snapshot);
status = transaction.completedEvidence(evidence);
```



*Alternative text: a stack of graphite request cards and a separate receipt stack enter an abstract policy box, which fills five blank result cells in a row for the replay ledger.*

The canonical sketch records only baseline cleanup confirmation, parasite rejection, one reset/presence transaction and its terminal release, receipt-provenance predictions, one strictly-post-deadline timeout and rollback confirmation, and shutdown confirmation. Its named result cells do not claim to replay addressed reads, Search ROM, or malformed receipts.

The host proof matrix separately covers every timing neighbor; presence absent, valid, early, late, and stuck; both copied read values; least-significant-first assembly; first, middle, and final slot failures; fieldwise changed duplicates; crossed correlation; single-drop restrictions; bounded search branches and collision; parasite rejection; timeout; cancellation; reset and shutdown cleanup; rollover and exhaustion; and two independent policies producing byte-stable results from the same trace.

## 10 Inspect gates before making a claim

| Gate      | E0 evidence                                    | Still open                                        |
|-----------|------------------------------------------------|---------------------------------------------------|
| resources | zero hardware owners and exact compiler report | physical pin/timer/interrupt ownership            |
| protocol  | typed phases and copied timing boundaries      | measured pulse, sample, rise, and recovery timing |
| cleanup   | correlated Release intent and receipt          | physical high-impedance/idle voltage              |
| identity  | byte-exact nonzero ROM transport               | family, CRC, specimen authenticity                |
| power     | parasite requests reject atomically            | any strong-pull topology or campaign              |

The ordinary Mega replay measures 10,390 bytes of flash and 735 bytes of static SRAM. The exact isolated no-LTO probe measures 12,116 bytes of flash, 735 bytes of static SRAM, and 199 bytes of conservative synchronous stack. The policy object is 254 bytes: an independently accepted miss above the 192-byte target, subject to a zero-growth constraint, and only two bytes below the 256-byte hard limit. Caller-owned request, search, intent, receipt, and snapshot values measure 29, 10, 45, 39, and 84 bytes, or 207 bytes together. After the specified reserve and stack accounting, 7,130 bytes of Mega SRAM remain.

The ordinary sketch and exact isolated probe are separate measurement boundaries; their flash and static-SRAM values must not be added. These compiler measurements do not establish electrical timing, physical cleanup, or sensor behavior. Independent review accepts both flash target misses and the object target miss; every hard and residual gate passes. The stable evidence fingerprint is e2a588c35c0e304c91aaf23ef4914f3f0b440afd3e9d37c87b171635ddb5e9bc.

**E1a remains open.** Before any powered adapter, qualify an exact externally powered specimen and ordinary external pull-up. Record package and pinout, genuine/compatible evidence, supply and current, drive-low and release rails, rise time, reset/presence/slot waveforms, interrupt effects, timeout cleanup, stuck-low behavior, startup/reset/shutdown, and an independent physical line-release observation. Parasite power remains E1b-closed.

## 11 Completion record

| Check                                                             | Result or evidence |
|-------------------------------------------------------------------|--------------------|
| Closed typed operation and nonzero-ROM transport boundary         |                    |
| Every configured microsecond-window neighbor and rollover         |                    |
| Intent/receipt correlation, duplicate, regression, and exhaustion |                    |
| Least-significant-first slots and attributable partial prefix     |                    |
| Timeout, cancellation, reset, shutdown, and cleanup confirmation  |                    |
| Parasite rejection and absence of hidden polling/retry            |                    |
| Canonical replay, exact resource evidence, and open E1 gates      |                    |

**Completion statement:** I can distinguish copied protocol evidence from electrical execution, correlate every bounded phase, preserve partial prefix and cleanup debt, explain why release is not proof of an idle bus, and name the physical evidence still required.

Learner: \_\_\_\_\_ Reviewer: \_\_\_\_\_ Date: \_\_\_\_\_

## Primary references

- Analog Devices, *DS18B20 Programmable Resolution 1-Wire Digital Thermometer*, Rev. 6, manufacturer data-sheet PDF.
- Analog Devices, *1-Wire Search Algorithm*, Application Note 187: manufacturer application-note page.
- Analog Devices, *1-Wire Extended Network Standard*, including its discussion of Application Note 126: manufacturer design-note page.

This print edition is untagged. Its text is searchable and copyable, and the HTML lesson provides the hardware-independent alternative. No PDF/UA, WCAG, or other PDF accessibility-conformance claim is made.